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Highly Luminescent and Stable Green Quasi-2D Perovskite-Embedded Polymer Sheets by Inkjet Printing

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Perovskite materials serve as promising candidates for display and lighting due to their excellent optical properties, including tunable bandgaps and efficient luminescence. However, their efficiency and stability must be improved for further application. In this work, quasi-two-dimensional (quasi-2D) perovskites embedded in different polymers are prepared by inkjet printing to construct any luminescent patterns/pictures on the polymer substrates. The optimized quantum yield reaches over 65% by polyvinyl-chloride-based quasi-2D perovskite composites. In addition, as-fabricated perovskite–polymer composites with patterns show excellent resistance to abrasion, moisture, light irradiation, and chemical erosion by various solvents. Both quantum yield and lifetime are superior to those reported to date. These achievements are attributed to the introduction of the PEA$^+$ cations to improve the luminance and stability of perovskite. This patterned composite can be useful for color-conversion films with low cost and large-scale fabrication.

1. Introduction

Hybrid organic–inorganic perovskites have been attractive for various optoelectronic applications in recent years, not only due to their excellent optical properties but also easy fabrication.[1–3] However, the stability of perovskites is always the obstacle that cannot be neglected. It is known that perovskites are extremely sensitive to water, oxygen, and UV light, which may result in quick degradation of perovskites and decreasing of luminescence.[4–6] Quasi-two-dimensional (quasi-2D) perovskites are formed by doping 2D perovskites into three-dimensional (3D) perovskites. Typically, quasi-2D perovskites can be described with the formula $R_2(ABX_3)_{n-1}BX_4$, where R is a bulky organic cation (such as an aliphatic or aromatic alkylammonium), A refers to smallish cation (for example, methylammonium or phenylethylammonium), B denotes a metal cation (Pb^{2+} or Sn^{2+}), X is a halide ion (Cl^-, Br^- or I^-), and n refers to the number of inorganic layers stacked together.[7,8] Recent works have verified that the highly efficient luminescence of quasi-2D perovskites is caused by formation of multiple quantum wells.[9–12] For example, Yang et al. found that the $PEA_2(MAPbBr_3)_2PbBr_4$ (n = 3) delivered the best photoluminescence (PL) among quasi-2D perovskites.[13] Furthermore, Zheng et al. demonstrated that quasi-2D perovskites presented better water stability in contrast to 3D perovskites thanks to the hydrophobic nature of the R (ammonium salts) cation.[14] Many researchers reported some 3D perovskite composite film encapsulated in polymer. The composites also showed greatly enhanced resistance to water.[15,16] This is because the fabricated perovskites were encapsulated by polymers, preventing the perovskites away from water erosion. However, once water and oxygen permeate through the polymer, the perovskite film would degrade very fast.[16,17] Therefore, the inherently stable quasi-2D perovskites exhibit huge potential for stable and highly efficient optoelectronic applications.

The fabrication of optoelectronic devices still requires efficient and effective patterning process for full-color display, color painting, and security label. So far, there are various techniques to pattern perovskite materials, such as mask-based lithography,[18–22] nanoimprinting,[23–25] and inkjet printing.[26–29] However, the UV-exposed processes in lithography may lead to degradation of perovskites.[8,22] For nanoimprinting, one could not avoid the complicated processes and direct contact with substrates.[25] Among these methods, inkjet printing technology is based on the ejection of droplets from nozzles, and then precise position of very small volumes of droplets on a substrate.[31] Therefore, inkjet printing is particularly attractive for its advantages like no-contact process, direct-writing fabrication

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of patterns, mask-free, and flexible substrate capability.[31–33] Currently, many researchers have successfully applied inkjet printing to fabricate solution processed electronic, such as conductive circuits, solar cell devices, and wearable electronic devices.[33] Very recently, Kim et al. have fabricated patterned $MAPbBr_3$ inside a blade coated polymer film by inkjet printing.[26] However, the fabricated film did not exhibit outstanding stability and the thin films could be easily damaged by scratching or some external forces. Therefore, exploring a way to prepare stable patterned perovskite–polymer composites could provide a viable solution for the practical application of perovskites in devices.

In this work, we proposed a facile in situ strategy using three commercially available polymers to achieve a patterned, stable, and highly luminescent quasi-2D perovskite–polymer composites by inkjet printing. This process yields composites with high PL quantum yield (PLQY) of over 65% and a narrow full width at half-maximum (FWHM) of 22 nm. The quasi-2D perovskite–polymer composites show great resistance against light, water, and air. The PL intensity of prepared composites still maintains 80%, when emerged in water for 20 days and exposed in air for 50 days, respectively. When irradiated with intense ultraviolet for 240 h, the PL still keeps 50% in intensity. This work confirms the effectiveness and the feasibility of making highly luminescent and stable perovskite–polymer composites by inkjet printing. This approach for stable and efficient perovskite composite patterning is worth special attention in display field.

2. Result and Discussion

Figure 1a demonstrates the preparation process of patterned quasi-2D perovskite composite sheets. The perovskite patterns were formed by inkjet printing on the polymer sheets. The precursor inks were prepared by adding phenethylammonium bromide (PEABr), methylammonium bromide (MABr), and lead bromide ($PbBr_2$) in *N*,*N*-dimethylformamide (DMF) solvent, while there was no PEABr in the controlled inks. The polymer was partially dissolved and swelled as soon as the precursor droplet was dropped on the surface of the polymer, due to the existence of DMF. The process of integrating the ink to the polymer is similar with swelling–deswelling method by spin coating.[15] The printed patterns were annealed to evaporate the solvent, and the quasi-2D perovskite microarrays were formed. The polymer sheets including polyvinyl chloride (PVC), polymethylmethacrylate (PMMA), and polycarbonate (PC) were chosen to demonstrate the generality of this strategy. The quasi-2D perovskites were fabricated successfully in all polymer sheets which can be observed through their luminescence under UV excitation. The microarrays on the surface of PVC, PC, and PMMA were shown in Figure 1b,c, respectively. The samples as-fabricated were denoted as PEA–PC, PEA–PC, and PEA–PMMA, respectively. Obviously, the dots in PEA–PMMA sample present an obvious coffee-ring effect because of the high Hansen solubility parameter of PMMA (up to 10.5), indicating that the ink tends to dissolve the PMMA.[34] In contrast, the dots on the PVC and PC exhibit higher uniformity. As shown in the scanning electron microscopy (SEM) of the dots on PEA–PVC and PEA–PC samples (Figure S1, Supporting Information), the perovskite nanocrystals cannot be directly observed on the surface, suggesting that the fluorescent perovskite was mainly embedded inside the polymer. The relation of the droplet diameter (*D*) and the contact angle (θ) of the droplet can be written as Equation (1). It has been simulated that *D* decreases as θ increases under the constant volume (*V*) of drops by using Matlab software (Figure S2, Supporting Information).[27,36]

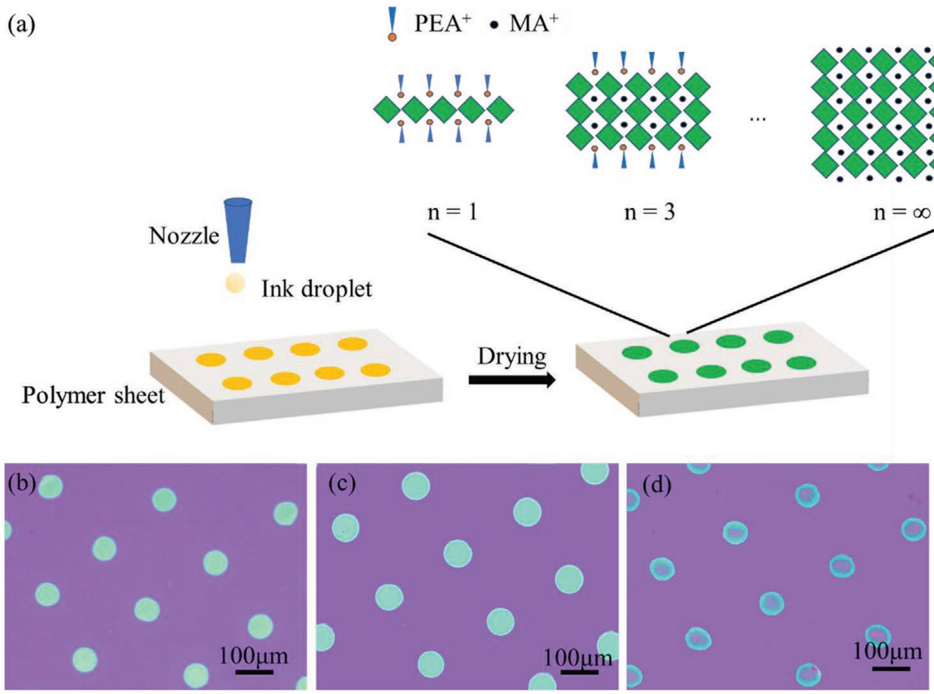

Figure 1. a) Schematic illustration of the preparation of quasi-two-dimensional (quasi-2D) perovskite composite sheets by inkjet printing. The images of printed patterns on different polymer sheets: b) polyvinyl chloride (PVC), c) polycarbonate (PC), and d) polymethylmethacrylate (PMMA), respectively.

Therefore, the D of these dots on PVC is the smallest (≈50 µm) because of the highest contact angle of the inks onto PVC (Figure S3, Supporting Information):

$$D = \sqrt[3]{\frac{24V\sin^3\theta}{2\pi - 3\pi\cos\theta + \pi\cos^3\theta}} \quad \left(0 < \theta \le \frac{\pi}{2}\right) \tag{1}$$

Optical properties of quasi-2D perovskite on different polymer sheets (10 × 10 mm) were measured by 100 × 100 dot array with a dot-to-dot pitch of 100 µm. The PLQY of perovskites within different polymers prepared by using various precursor concentrations are summarized in Table S1, Supporting Information. As shown in Table S1 (Supporting Information), the quasi-2D based composites show higher PLQY than MA-based composites, due to the formation of multiple quantum wells in quasi-2D perovskite. Significantly, a PLQY of 65% is achieved by using PEABr as precursor and PVC as printing substrate, which is higher than other composites. Additionally, PEA–PVC, PEA–PMMA, and PEA–PC samples show similar absorption and PL spectra except the PL intensity (Figure 1a,b). The emission peaks in the range 515–517 nm correspond well with the previous report.[13,35–37] We noticed a slight blue shift of 2 nm for PEA–PMMA samples which probably resulting from the small swelling ratio and high dispersibility of PMMA in DMF.[15,38] In addition to the absorption edge around 520 nm for bulk phase, different phase composition in PEA–polymer lead to discrete absorption features located at 400, 425, and 450 nm, which correspond to the absorption of $n = 1$ (PEA_2PbBr_4), $n = 3$ ($PEA_2(MAPbBr_3)_2PbBr_4$), and $n = 5$ ($PEA_2(MAPbBr_3)_4PbBr_4$) phase, respectively.[13] Besides, the PEA–PVC sample demonstrated the highest PL intensity compared to PEA–PC and PEA–PMMA, which also possesses the highest PLQY among three (PLQY = 65%).

As already explained by other literatures, the higher PLQY of PEA–PVC compared with other samples infers chlorine atoms in PVC assist to passivate defects in perovskite.[16,35] Moreover, the chlorine atoms may affect the crystallinity of perovskite to further enhance its performance.[39] It is worth noting that this is one of the highest values reported for patterned composites. **Figure 2**c shows the photograph of as-printed PEA–PVC sample under UV radiation which exhibits uniformly bright green luminescence. The inset is the same sample under sunlight for comparison. The absorption and PL spectra (Figure 2d) for PEA–PVC and MA–PVC are consistent with the reported results.[13,37] The results show that formation of quasi-2D perovskite is mainly affected by PEA precursor but independent of polymers. Moreover, both samples demonstrate good transparency within visible region (Figure 2e) compared with bare PVC, which may particularly give potential to inkjet printing of perovskite in transparent display.

The time-resolved PL (TRPL) measurement was performed to investigate the difference of PLQY between MA–PVC and PEA–PVC composite sheets (Figure 2f). The PEA–PVC sample shows longer average PL lifetime than MA–PVC sample, which can be ascribed to the energy fueling in multiphase perovskites.[41] Specifically, ultrafast charge transfer occurs from small-n phases to bulk phase which reduces the possibility of nonradiative exciton quenching. Consequently, the PL for PEA–PVC is enhanced compared to that of MA–PVC (Figure 2a). Energy transfer processes in PEA–PVC were verified by transient absorption (TA) spectroscopy. As shown in Figure 2e, the TA spectra for PEA–PVC sample acquired at different delay times show the absorption bleaching for $n = 1$ (PB_1, ≈400 nm), $n = 3$ (PB_2, ≈434 nm), and bulk phase (PB_3, ≈500 nm), which is consistent with the absorption spectrum. Figure 2f shows the normalized decays at selected probe wavelength. The PB_2 ($n = 3$; 2D phase) shows an ultrafast decay, while PB_3 ($n = \infty$; bulk phase) starts to decay after keeping a maximum intensity for several picoseconds. These results indicate the occurrence of energy transfer from PB_2 to PB_3 phases.[40]

The stability of perovskites is a well-known challenge and the moisture and light irradiation can seriously accelerate their degradation. The inkjet printed PEA–PVC and MA–PVC samples with high PLQY were aged to test the stability under various conditions. The photostability was tested at ambient conditions under irradiation by a fluorimeter lamp offering an excitation wavelength of 400 nm and light intensity of 25 mW cm^{-2}. As shown in **Figure 3**a, after exposed to irradiation for 240 h, the PL intensity of MA–PVC and PEA–PVC samples maintain about 50% of their initial values. This indicates that the micropatterns fabricated by this method demonstrate good stability against light irradiation. Because the polymer insulate perovskite from the oxygen, which prevent them from reaction under light irradiation. The narrow FWHM of PL suggests a good distribution of perovskites in MA–PVC (≈21 nm) and PEA–PVC (≈22 nm). After 330 h irradiation, the PEA–PVC exhibited a FWHM of 30 nm, but the FWHM of MA–PVC is near 40 nm. Furthermore, Figure 3b illustrates the fluctuation of the PL intensity of PEA–PVC and MA–PVC exposed in air. The PL performance of PEA–PVC is very stable in air condition, since it maintained almost 80% PL intensity compared to the initial value after 50 days. However, the PL intensity of MA–PVC sample decreased 50% implying that $MAPbBr_3$ presents a worse stability in contrast to the quasi-2D perovskite. As Figure 3c shows, when the samples are exposed to water, the PL intensity of MA–PVC sample decreases dramatically, and the PL intensity is down to 40% in contrast with initial values after 50 days. However, the PL intensity of PEA–PVC sample only dropped by 30%. Based on the tests of degradation, we believe that the quasi-2D perovskite plays an important role in enhancement of stability. First, PEA^+ cations in the quasi-2D perovskite helps to improve the air and water stability due to its hydrophobicity. Second, the degradation of perovskite is likely to begin via active defects.[41] The aliphatic or aromatic alkylammonium cations like PEA^+ or BA^+ can improve the overall crystallinity to reduce the number of defects,[42] which contribute to the enhanced long-term stability. Third, the air and water damage the 3D perovskite through the grain boundaries.[43] In quasi-2D perovskite, the 2D phase at the grain boundaries is likely to suppress the degradation in the grain boundaries. Therefore, the PEA–PVC sample displays an overall better stability against air, water, and light irradiation compared with MA–PVC samples. In addition, because of the good chemical resistance of PVC,[44–45] the PEA–PVC sample is also proved to be stable in acid, alkali, and ethanol (Figure 3d), so that it can be directly used in many extreme conditions during optoelectronics processing.

With preprogrammed inkjet printer, a large patterned polymer composite sheets can be fabricated. **Figure 4**a shows

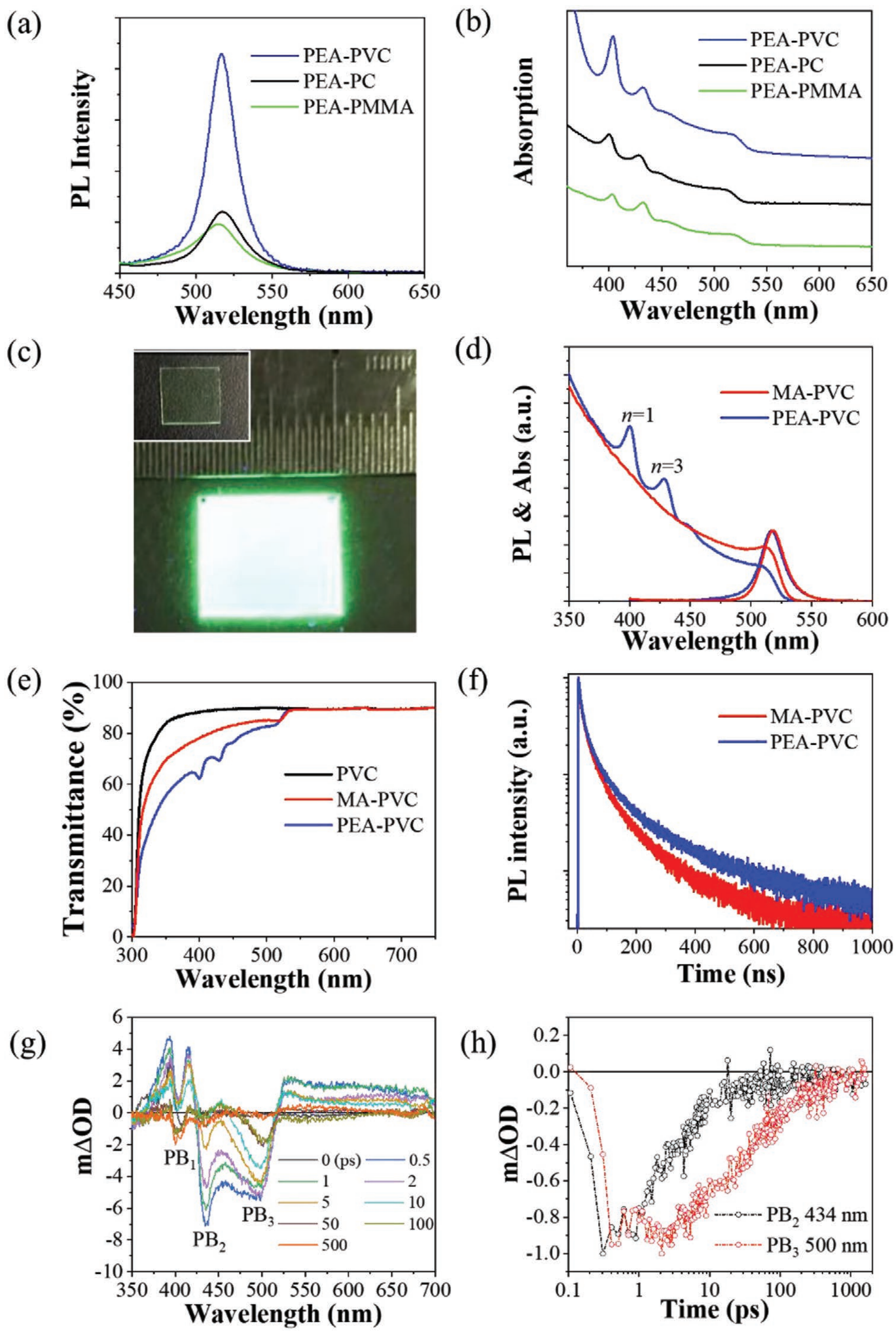

Figure 2. a) Photoluminescence (PL) and b) absorption spectra of PEA-polymer sheets. c) The image of PVC–PEA (where PVC is polyvinyl chloride) sample under UV lamp. d) The PL and absorption spectra of PVC–MA (where MA is methylammonium) and PVC–PEA. e) The transmittance spectra of PVC, PVC–MA, and PVC–PEA. f) PL decays of as-prepared samples excited at 405 nm and monitored at 510 nm. g) Transient absorption (TA) spectra of PVC–PEA at different delta times with 400 nm excitation. h) Normalized TA decay traces of $n = 3$ (PB_2, 434 nm) phase and bulk phase (PB_3, 500 nm).

the image of quasi-2D perovskite patterned samples under UV lamp. The red emission came from the $MAPbI_x/Br_{3-x}$, whose PL spectrum is shown in the inset of Figure 4a. The micrograph of the red area on Figure 4a is enlarged in Figure 4b and the arrow represents the moving direction of the nozzle, which suggests the as-printed perovskite dots are uniform and well-aligned. As Figure 4c depicts, the microdots show a concave surface with the depth of 250 nm appeared on the surface of PVC after the solvent evaporates. Additionally, the patterned samples are also wear-proof to resist the scratching (video 1), which is indispensable

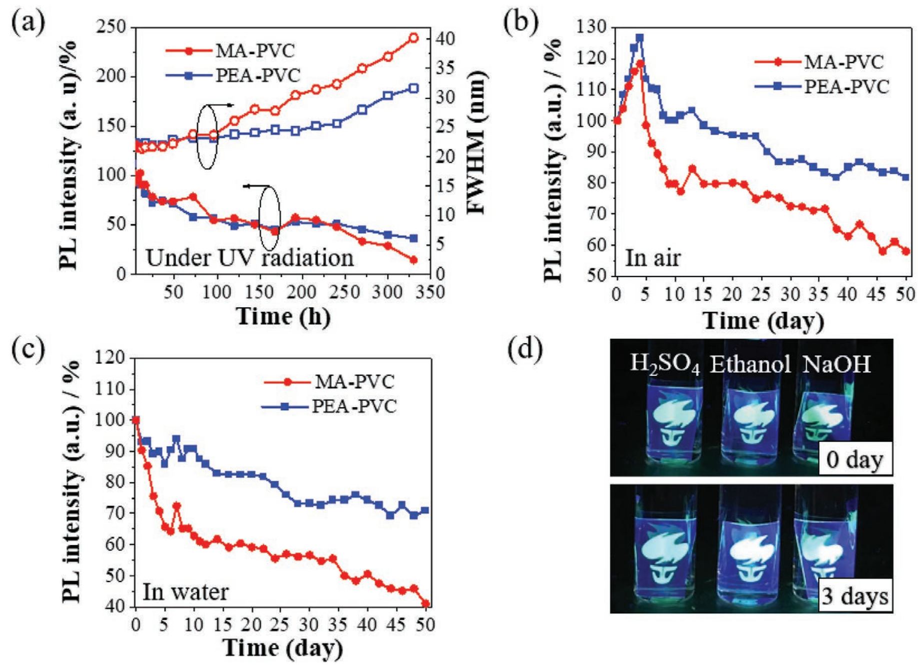

Figure 3. Stability tests of perovskite–polymer composite sheets in various conditions. a) Change of the photoluminescence (PL) intensity and full width at half-maximum (FWHM) of MA–PVC and PEA–PVC samples under blue light radiation with 25 mW cm^{-2} for 325 h. b) Change of the PL intensity of MA–PVC and PEA–PVC samples in air for 50 days. c) Change of the PL intensity of MA–PVC and PEA–PVC samples in water for 50 days. d) PEA–PVC samples in various solvents. The composite sheets were immersed in each solvent for 3 days before taking photograph.

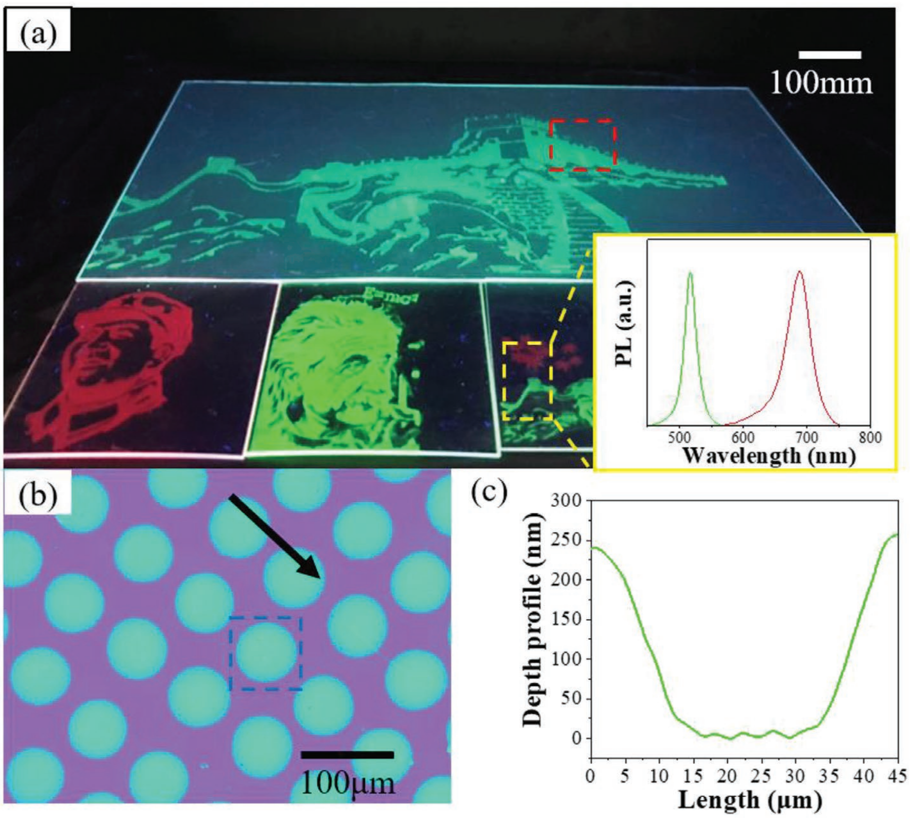

Figure 4. Application of perovskite–polymer composite sheets for large area picture. a) Photograph of patterned perovskite–polymer composite sheets under UV illumination. b) The microscopic fluorescent images of the red bulk in (a) and the black arrow shows the moving direction of the nozzle. c) The depth profile of the microdots, the scan area is marked in blue (b).

quality for applications. The combination of large area, low cost, high stability, high luminescence efficiency of quasi-2D material, and patternable properties make this strategy promising for various potential applications.

3. Conclusion

In conclusion, we have fabricated quasi-2D perovskite embedded in polymers with the capability of patterning by inkjet printing successfully. In comparison with MA–polymer samples, the PEA–polymer exhibits significant improvement in PLQY (over 65%) due to the self-organized quantum-well structure. Additionally, PEA–PVC sample presents outstanding optical property and the printed patterns show better homogeneity, compared with samples using other polymers. That is probably because the chlorine atom in PVC affects the crystallinity and reduce the defects of perovskite. More importantly, because of the inherent stability of quasi-2D perovskite and the outstanding barrier property of PVC, the PEA-–PVC sample obtains excellent resistance to the abrasion, air, water, light irradiation, and various solutions. We anticipate that this approach holds promise for applications such as fluorescent advertising boards with a large area.

4. Experimental Section

Materials: PC sheets (Terry-Uttley company), PVC sheets (Terry-Uttley company), and PMMA sheets (Terry-Uttley company), Lead (II) bromide ($PbBr_2$ 98%, Sigma-–Aldrich), PbI_2 (lead(II) iodide 98.5%, Sigma–Aldrich), *N*,*N*-dimethylformamide (C_3H_7NO 99.8%, Sigma–Aldrich), CH_3NH_3Br (methylamine hydrobromide 98%, Xi'an p-OLED), and phenylethylammonium bromide (PEABr 98%, Xi'an p-OLED).

Patterned Films Fabrication: Quasi-2D perovskite inks, PEABr, MABr, and $PbBr_2$ (2:2:3 molar ratio) with concentration for $PbBr_2$ of 0.01 mol mL^{-1}, 0.02 mol mL^{-1}, and 0.05 mol mL^{-1} were prepared in DMF and stirring for 3 h. For 3D perovskite inks, the method only removed PEABr and the others were the same as quasi-2D perovskite inks.

The polymer substrates were cleaned by ultrasonication in deionized water, and then dried by using flowing nitrogen gas. The inkjet printing on the polymer substrates was performed by using a Microfab JETLAB 2 printer equipped with a piezoelectric-driven inkjet nozzle (diameter: 40 µm) in glovebox and the electric-driven of the inkjet printer was shown in Figure S3, Supporting Information. Finally, the solvent was removed rapidly at 60 °C in the glove box with the pressure of 2–3 mbar.

Characterization: The morphologies of the microarrays on polymer substrates were characterized by using a fluorescence microscope (INSIZE ISM-D500 equipped with UV-light of 400 nm). The SEM image was obtained by using Zeiss Gemini300. The contact angle was determined using an instrument (JC200C1,6 Powereach, China) based on ellipse-fitting at room temperature. The PLQY and PL intensity of the film was measured at by using the absolute PLQY spectrometer (Quantaurus-QY C11347-11) with an excitation of 350 nm at room temperature. UV–Vis absorption spectra were recorded on UV–violet spectrometer from Beijing Spectrum Analysis 1901 Series at room temperature. Lifetime of the samples were recorded by Edinburgh Instruments FS5 (excited at 365 nm and monitored at peak position for each sample) and the lifetimes were fitted by Fluoracle software. PL decays were measured by PicoQuant FluoTime 300 spectrometer. The samples were excited at 405 nm by a laser diode with a repetition rate of 1 MHz, and the decays were acquired by UV–VIS photon detector and TCSPC electronics (monitored at 510 nm). TA spectra were taken from a Newport Transient Absorption Spectrometer. The pump (400 nm) and probe pulses were generated by coupling the seed pulses from a regenerative amplifier (≈100 fs, 1 kHz) into an optical parametric amplifier (TOPAS, Light Conversion) and a CaF_2 plate, respectively. For irradiation stability test, the PEA–PVC and MA–PVC samples were irradiated with 25 mW cm^{-2} by using a high-power blue LED (400 nm). The PEA–PVC and MA–PVC samples were put in the air and immersed in de-ion water at the room temperature for 50 days, aiming to test the air and water stability of that samples.

Statistics: To obtain more reliable results, the experiments were performed using replicate samples (three or four samples) in each series. The average values and standard deviation were calculated from the numerical data.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

inkjet printing, lighting, perovskites, quasi-2D

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